LUMAFLOW · CALCULATION

LumaFlow sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-25	Amish Chadha	Kit 1.9.1: rerun on the detailed model (kit 1.9 interface; dimensions unchanged); after the independent review: 275 nm response factor (B10), LED derating stack (B11), weighted alarm threshold (C3 to C6, D-S4), outlet pressure sensor start-up (D5, E2, D-S3), Kv 0.10 valve case (G9); R2 and R8 at risk; cost USD 433
0.6	2026-10-02	Amish Chadha	Approved follow-ups carried out: section B and the window stress re-run for the 0.5 mm PTFE window washer (LED gap 1.3 mm); new C6 for the UV level bar (R18, now met); idle LED pulse in E4, E6 and F7; mass H2; cost with the washer, interlock plug, bar and labels (USD 416)
0.5	2026-10-02	Amish Chadha	Requirement table: R12 interlock wording and R18 added after the 2026-10-02 decisions (LMF-DEC-001); no figure changed. The window seat washer (A2, option b) is not yet re-run in section B
0.4	2026-10-01	Amish Chadha	

			Rerun for the constructable design (LMF-DDR-003); new G8 for the upper cap roof; cost reported against the value-engineering target
0.3	2026-09-26	Amish Chadha	Budget top-up approved by Amish; rerun with budget_usd $340, R16 now met
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); rerun for the 50 mm bore, 0.95 liner, 1.2 L/min, wall dose sensor, 15 Hz per L/min flow sensor, 10 mm window, M5 rods, pressure limiter and thermal cut-back
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (flow, Monte Carlo optics and dose, dose monitor, switching, power, thermal, pressure drop, window and structure, size, cost, options)

LumaFlow sizing calculations

This note checks the LumaFlow design as revised by LMF-DDR-002 and made constructable by LMF-DDR-003: a 50 mm bore lined with high-reflectance PTFE, a design flow of 1.2 L/min (0.32 gpm), a UV-C photodiode in the tube wall at mid height, a flow sensor of 15 Hz per L/min, an upstream pressure limiter and a thermal cut-back. On paper, LumaFlow meets twelve of its eighteen requirements, has three at risk (R2, R4, R8) and misses one (R3); one cannot be verified at TRL 3, and the cost (R16) is over its value-engineering target. The reduction equivalent dose (RED) in clear water (90 %/cm UV transmittance, UVT) is 50.9 (laminar) to 74.9 (plug) mJ/cm² for a 254 nm-equivalent response, and 47.3 mJ/cm² with the published response at 275 to 280 nm [B10], so **R2 is at risk**: above 40, but short of the 20 % margin this note requires (v0.7, after the independent review of 2026-10-25); in v0.1, with a 25 mm bore at 2.0 L/min, it was 14.7 to 19.0 mJ/cm². The dose at the 70 %/cm test condition is 17.0 to 21.3 mJ/cm² (**R3 not met**, kept visible by decision D1). Value-engineering target: USD 340. Estimated cost of the constructable design: USD 433 (USD 93 over the target) [I1]. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [B5], is the line of that script's output that carries it.

> **Safety:** These calculations concern a device meant to disinfect drinking water with UV-C light under mains pressure. They are first-principles estimates for a paper proof of concept, not a substitute for biodosimetry, datasheets or a review by a qualified engineer. A calculated dose above target is not a demonstrated dose. LumaFlow is a research and educational prototype, not a certified water treatment device. See LMF-PRC-001, Safety.

Scope and method

The note checks every requirement in LMF-REQ-001 v0.8 against the design in LMF-PRC-001 v0.8 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `levels()` and part volumes, so the dimensions used here are the ones in the STEP files and in drawing LMF-DWG-001 Rev P5. It reads prices from `bom/bom.csv` and the value-engineering target (`budget_usd`) from `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py` (about 20 s; it runs the ray trace eight times).

Status rule for the dose: the laminar-flow RED is the design value, because the laminar profile is the more pessimistic of the two limits and the real profile lies between them [A5]. **Met** needs the laminar RED to clear the target by 20 %.

Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Flow	1.2 L/min (0.32 gpm), set by the restrictor; water at 20 °C for hydraulics, 25 °C for the thermal case	LMF-REQ-001 v0.4, R1
LEDs	Six 275 nm LEDs on a 32 mm circle, 60 mW UV-C each at 350 mA and 6.2 V (0.36 W, 13.02 W electrical); Lambertian emission	Typical 3535-class UV-C LED ratings; bin to be confirmed
Window	UV-grade fused silica, 57 x 10 mm on a 51 mm seat, n = 1.496 at 275 nm, bulk transmittance 0.995; air gap 1.3 mm (the window sits on a 0.5 mm PTFE washer on the retaining ring); unpolarized Fresnel losses at both faces; light outside the seat aperture or the 50 mm bore is lost	Standard optics
Water	n = 1.372; base-e absorption α = −ln(UVT) per cm (0.105/cm at 90 %/cm, 0.357/cm at 70 %/cm); no scattering	UVT definition
Walls	Diffuse (Lambertian) reflection. Wetted high-reflectance PTFE liner 0.95 (sensitivity 0.80 and 0.90); wetted 316 stainless in the lower cap bore 0.30; window and LED side seen from the water 0.10	Engineering judgment for a sintered or expanded high-reflectance PTFE; to be measured
Ray trace	600,000 rays per case, 20 equal-area annuli by 60 axial bins; fluence rate = absorbed power / (α x bin volume); the wall sensor window is an 8 x 8 mm absorbing patch at mid height	Collision estimator
Flow profile	Two limits: plug flow and fully developed laminar (parabolic) flow	Entrance length [A5]
Dose response	First-order inactivation of a challenge organism with 20 mJ/cm² per log (MS2-like), with the same response at 275 nm as at 254 nm	Assumption; the wavelength response must be checked
Photodiode	SiC, 0.13 A/W at 275 nm, 0.06 mm² active area, behind an 8 mm quartz window of 0.92 transmittance in the tube wall	Typical TO-46 SiC photodiode
Flow sensor	Hall-effect turbine, 15 Hz per L/min; Kv 0.35 m³/h	Pulse rate set by LMF-DDR-002; Kv to confirm
Valve	24 V DC normally closed, 4.8 W (0.2 A) held open, Kv 0.25 m³/h	Typical small direct-acting solenoid
Electronics	Driver 90 %; controller 0.6 W active; adapter 88 % at load, 70 % at light load, 0.1 W no-load	US DOE Level VI no-load class for small adapters
Use	15 L/day in 20 draws, 5 s run-on per draw	LMF-REQ-001
Thermal	Sink natural convection 5 W/(m²·K) (fins hanging down) plus radiation, ε = 0.85; 0.5 mm pads of 3 W/(m·K); 316 at 16 W/(m·K); water film 300 W/(m²·K) in the lower cap (range 150 to 600); 10 K/W junction to board per LED	Engineering judgment; film coefficient is the main uncertainty
Structure	Fused quartz design tensile stress 6.8 MPa (1,000 psi) (Momentive, mechanical properties); Poisson ratio 0.17; simply supported edge; 8 bar working pressure; pressure limiter set at 4 bar	Roark circular plate, uniform load

A. Flow and hydraulics (R1)

The irradiated channel is 50 mm across and 242 mm long, from the window to the PTFE top disc; the flow path from inlet to outlet is 226 mm [A1]. The channel holds 474 mL, 444 mL of it between the ports [A2]. At 1.2 L/min the mean velocity is 1.02 cm/s and the mean residence time 22.2 s [A3]. The Reynolds number is 507 at 20 °C and 334 at 5 °C, so the flow is laminar [A4]. The laminar entrance length, about 1.3 m, is far longer than the flow path [A5], so the velocity profile lies between plug flow and a parabola; both limits are carried through section B. R1 is met by the 1.2 L/min restrictor, and the flow sensor range of 0.3 to 6 L/min covers it.

B. Optics and UV-C dose (R2, R3)

The six LEDs give 0.36 W of UV-C from 13.02 W electrical, a UV-C efficiency of 2.8 % [B1]. Of that, 84.8 % enters the water [B2]; it was 87.0 % with the 0.8 mm gap of v0.5, because the LEDs now sit 0.5 mm lower and a little more of their light misses the 51 mm open span. In the 50 mm bore light crosses twice as much water between wall reflections as in the old 25 mm bore, and the liner returns 95 % of what reaches it. At 90 %/cm the water absorbs 52.9 % of the LED output and the walls 25.9 % (v0.1: 26.1 % and 49.8 %); at 70 %/cm the water absorbs 66.5 % [B3, B4].

Table 2. Dose at 1.2 L/min (mJ/cm²). Average is the flow-weighted average dose; RED is for the 20 mJ/cm² per log organism.

UVT (%/cm)	Average dose	RED, plug flow	RED, laminar flow	Reactor efficiency, laminar
95	131.9	131.4	82.7	0.63
90	75.4	74.9	50.9	0.67
85	50.8	50.5	35.9	0.71
80	36.8	36.5	27.2	0.74
75	27.8	27.5	21.3	0.77
70	21.5	21.3	17.0	0.79

At the design water quality of 90 %/cm the RED is 50.9 (laminar) to 74.9 (plug) mJ/cm² [B5]. The laminar value is 27 % above the 40 mJ/cm² target for a 254 nm-equivalent response; with the response factor of 0.93 for 275 nm it is 47.3 mJ/cm², 18 % above [B10], so **R2 is at risk** under this note's 20 % rule. A worst-case stack of LED derating (about 12 % less output at the junction temperature, a 10 % low bin) gives 37.5 mJ/cm² [B11], to be checked on the chosen LED. At 70 %/cm it is 17.0 to 21.3 mJ/cm² [B6], 15.8 with the 275 nm factor, so **R3 is not met**, as decision D1 accepted.

The reactor efficiency (RED over average dose) is 0.67, lower than the 0.77 of v0.1: the slow laminar core on a wide bore sees less light than the water beside the wall. Streamline doses at 90 %/cm range from 41 mJ/cm² on the axis to 79 at mid radius and 1,433 beside the wall [B7]. The liner reflectance is the main optical uncertainty: 45.2, 48.7 and 50.9 mJ/cm² at 0.80 (plain PTFE), 0.90 and 0.95 [B8], so even plain PTFE meets 40 mJ/cm² but not the 20 % margin. LEDs 20 % below rating (bin, aging or temperature) give 40.7 mJ/cm² [B9].

C. Dose monitor (R4)

The photodiode now sits behind an 8 mm quartz window in the tube wall at mid height (Z = 166 mm), facing the enclosure. It sees 0.19 mW/cm² and gives about 13.4 nA at 90 %/cm, 2.4 nA at 80 %/cm and 0.19 nA at 70 %/cm [C1], a signal across the whole range; the top-disc position of v0.1 gave 2.4 nA at 90 %/cm and nothing resolvable at 70 %/cm. Between 90 and 85 %/cm the wall signal falls 2.6 times as fast, in log terms, as the RED [C2] (4.4 times for the old position).

Table 3. Wall sensor signal against UVT.

UVT (%/cm)	Irradiance (mW/cm²)	Photocurrent (nA)	Signal relative to 90 %	RED relative to 90 %
95	0.752	54.0	4.02	1.63
90	0.187	13.4	1.00	1.00
85	0.077	5.5	0.41	0.71
80	0.033	2.4	0.18	0.54
75	0.013	0.96	0.072	0.42
70	0.003	0.19	0.014	0.34

The controller keeps the proportional mapping RED_est = RED_ref x (S/S_ref) x (Q_ref/Q), which blames every loss of signal on the LEDs and so always under-reads. With RED_ref = 47.3 mJ/cm², the 275 nm weighted value (STEM fix D-S4, 2026-10-25; v0.6 used 50.9), the alarm threshold is S/S_ref = 0.85 [C3], so one failed LED of six (S/S_ref 0.83) trips it. With LEDs at rated output the alarm trips below about 89.1 %/cm UVT, and LEDs aged to 85 % of output trip it in 90 %/cm water, where the true weighted RED is 40.0 mJ/cm² [C4]. A mapping that blamed the water instead would be unsafe: with LEDs at 70 % output in 90 %/cm water it would claim 44.9 mJ/cm² when the true RED is 35.6 [C5]. R4 stays **at risk** because the sub-nanoampere signal in poor water, the mapping and the 1 s fail-safe still need firmware and a test, which are TRL 4 work. The wall-sensor signals in poor water come from few rays in the trace, so the sub-nanoampere figures move between runs (0.38 nA at 70 %/cm in v0.5, 0.19 nA now); the trend, not the last digit, is the result.

The five-segment UV level bar on the enclosure lid (R18) shows the same wall signal relative to the alarm threshold, not a dose. Its segments light at 1.0, 1.1, 1.2, 1.3 and 1.5 times the alarm signal. With rated LEDs it shows two of five segments in 90 %/cm water (1.18 times the alarm signal) and all five above about 90.9 %/cm; the last segment goes out at 89.1 %/cm, exactly where the alarm trips. In 90 %/cm water the segments go out one by one as the LEDs age to 127, 110, 101, 93 and 85 % of their rated output [C6]. **R18 is met** by design; the bench check against the wall sensor reading is TRL 4 work.

D. Flow switching and run-on (R5)

At the 0.3 L/min switch-on threshold the 15 Hz per L/min sensor gives 4.5 Hz, one pulse every 0.22 s [D1]. With the LED rise time the worst-case switch-on is 0.24 s against 0.5 s [D2] (v0.1: 0.46 s with a 7.5 Hz per L/min sensor): **R5 is met**.

During the 5 s run-on the water held in the channel receives a volume-average 19 mJ/cm² at 90 %/cm [D4]. Taking the dose each parcel collected while flowing plus the run-on, every parcel in the flow path has at least 61 mJ/cm² when flow stops (plug flow) [D3].

E. Power and energy (R9)

The LEDs draw 13.02 W; with the driver, the 4.8 W valve and the controller the 24 V bus carries 19.87 W, and the mains draw is 22.6 W while water flows [E1]. Standby is 0.23 W, dominated by the adapter's no-load draw, the powered flow sensor and the outlet pressure sensor that starts the unit when the faucet opens [E2, D5]. **R9 is met** (25 W and 0.5 W).

At 15 L/day and 1.2 L/min the LEDs run 14.2 min/day (86 h/year); the unit uses 10.4 Wh/day or 3.8 kWh/year [E3]. The idle pulse decided on 2026-10-02 runs the LEDs for about 10 s every 4 h of idle with the valve shut: 59 s a day, 6.0 h a year of LED time, 17.1 W while it runs and 0.28 Wh a day, so the unit uses 3.9 kWh a year in all [E6]. With the pulses the LEDs run 92 h a year, and 10,000 h of LED on-time would take 108 years (116 years without them) [E4], so LED life is still set by aging in storage and by switching, not by on-time. For comparison, tap-scale mercury units drawing 13 to 22 W all day (VIQUA VT1, VT4 and S2Q-PA spec sheet) use 114 to 193 kWh/year [E5].

F. Thermal (R10)

The LEDs turn 12.66 W into heat. The larger 90 x 90 mm finned sink rejects it to cabinet air through 2.46 K/W; the sink, spreader ring, lower cap and board store 780 J/K [F1]. The water path through the spreader ring and the 316 stainless lower cap is 1.58 K/W and carries 8.9 W of the 12.7 W into the water, which warms by 0.11 K [F2].

With continuous flow the LED board settles at 45.1, 40.5 or 37.3 °C for water-film coefficients of 150, 300 or 600 W/(m²·K) [F3], below 50 °C in every case; the LED junctions sit near 62 °C [F4]. The board temperature sensor and the cut-back of LED current above 50 °C (decision in LMF-DDR-002) are a backstop for the uncertain film coefficient: **R10 is met**. An acetal lower cap, with no water path, would settle at 63 °C and pass 50 °C after 30.5 min of flow [F5], which supports the 316 lower cap (LMF-DDR-002 confirms it). If the LEDs stuck on with no flow, the board would head for 53 °C, where the cut-back would act; the 5 s run-on adds only 0.08 K [F6]. A 10 s idle pulse in still water warms the LED head by 0.16 K [F7].

G. Pressure drop, window and structure (R7, R8, R17)

Table 4. Pressure drop at 1.2 L/min, excluding the flow restrictor [G1, G2].

Element	Drop (bar)
Flow sensor (Kv 0.35)	0.04
Solenoid valve (Kv 0.25)	0.08
1 m of 1/4 in bore tube	0.012
Fittings (K = 6 in all)	0.012
Reactor ports	0.002
Reactor channel	below 0.00001
Total	**0.15 (2.2 psi)**

R8 is now worded to exclude the flow restrictor, which absorbs excess supply pressure by design. The unit drops 0.15 bar against 0.5 bar [G2]; with a Kv 0.5 sensor and a Kv 0.4 valve it would be 0.08 bar, and at the minimum supply of 2 bar, 1.85 bar remains for the restrictor and faucet [G3]. The margin covers the assumed flow coefficients, so **R8 is met**; the coefficients are still to be confirmed from the datasheets of the parts chosen.

The 57 x 10 mm window on its 51 mm seat carries a peak stress of 6.18 MPa at 8 bar, inside the 6.8 MPa design stress [G4]: **R7 is met**, with a 9 % margin (v0.1: 4.46 MPa on a 26 mm seat). Re-run for the PTFE washer under the window: the washer's 51 mm bore matches the retaining ring, so the open span and the stress are unchanged; a compliant washer makes the simply supported edge of the calculation a closer match to the real seat than metal on glass was. Unprotected, a 16 bar water-hammer spike would raise it to 12.4 MPa. With the pressure limiter set at 4 bar the window carries 3.09 MPa static, and a transient doubling to 8 bar gives 6.18 MPa [G5]: **R17 is met** as an installation requirement. The 8 mm sensor window sees 1.69 MPa and the tube hoop stress is 8.3 MPa [G6]. Each cap carries an end load of 2,655 N; four M5 316 studs, threaded 12 mm into the stainless lower cap, take 664 N each, 47 MPa in the thread [G7]. The acetal upper cap closes the channel above the PTFE top disc; its roof over the 56.2 mm liner pocket is 13 mm thick and carries 2.8 MPa at 8 bar with a clamped edge, against a long-term design stress of 10 MPa taken for acetal [G8]. The 3 mm roof of the concept model would have carried about 59 MPa, near the short-term strength of acetal, which is why LMF-DDR-003 made the cap 40 mm tall.

H. Size and I. Cost (R14, R16)

The unit is 277 x 100 x 322 mm without the adapter, inside the 350 x 150 x 350 mm envelope [H1]: **R14 is met**. It weighs about 5.0 kg dry and 5.4 kg full of water; the aluminium bracket plate (1.05 kg), the stainless tube (0.98 kg) and the stainless lower cap (0.94 kg) are the heaviest parts, and the adapter adds about 0.2 kg on the floor [H2]. Made parts are weighed from the model's volumes; bought parts at typical catalogue masses.

Value-engineering target: USD 340 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 433 (USD 93 over the target), from the 19-line BOM [I1]. The largest lines are the LEDs ($54), the machined 316 lower cap ($48) and the high-reflectance PTFE liner ($45) [I2]. The concept BOM was $338.00; making the design buildable (LMF-DDR-003) added the welded sensor boss, the window retaining ring and seals, the threaded ports with stem adaptors, the drilled bracket plate and saddles, the pipe clamps and more fixings (USD 402 in v0.5). The decisions of 2026-10-02 add USD 14: the PTFE window washer (USD 2), the LED head plug, socket and interlock-loop cable (USD 4), the five-segment UV level bar (USD 3) and the labels line (USD 5). R16 is reported against the target, not as a failure; savings worth trying are listed in the design decisions register (LMF-DEC-001).

J. Flow limits and options

Table 5. Flow limits for this reactor and the options against R3.

Case	Result
Six LEDs, 90 %/cm	40 mJ/cm² (laminar) up to 1.56 L/min; 48 mJ/cm² (Met with margin) up to 1.26 L/min [J1]
Same reactor at the former 2.0 L/min	32.1 (laminar) to 45.0 (plug) mJ/cm² [J2]
Option B (R3 at 70 %/cm)	15 LEDs, 33 W of LED power [J3]; R9 would fail
Option C (R3 at 70 %/cm)	Six LEDs meet 40 mJ/cm² below 0.46 L/min [J4]; proportional valve, later upgrade under D1

The 1.2 L/min design flow sits inside the 1.26 L/min limit for the 20 % margin; the 0.5 mm washer took 0.04 L/min off that limit. Filling a 1 L bottle takes about 50 s.

K. Requirement status

Table 6. Requirement status at TRL 3 (LMF-REQ-001 v0.8). Not met items first.

ID	Target	Value (tag)	Status
R3	RED 40 mJ/cm² or more at 1.2 L/min, 70 %/cm	17.0 to 21.3 mJ/cm² [B6]	**Not met** (accepted by D1)
R16	Value-engineering target USD 340; no custom PCB	$433.00 [I1]; module-based electronics	Over the target by USD 93
R4	Dose monitor; alarm and valve closed within 1 s	Wall signal 13.4 nA at 90 %/cm, 0.19 nA at 70 %/cm; alarm at S/S_ref 0.85, below about 89 %/cm or on one failed LED [C1, C3, C4]	At risk
R15	Window and LED head replaced in 15 min	LED head off on four screws from below, window out after six ring screws, no plumbing disturbed; needs a build to time	Not verifiable at TRL 3
R1	1.2 L/min, restrictor	Restrictor; sensor range covers it [A3]	Met
R2	RED 40 mJ/cm² or more at 1.2 L/min, 90 %/cm	50.9 (laminar) to 74.9 (plug) mJ/cm² [B5]; 47.3 weighted for 275 nm [B10]	At risk
R5	LEDs on within 0.5 s above 0.3 L/min; 5 s run-on	Outlet pressure sensor wakes the unit, LEDs on before the valve opens; flow confirmed in 0.24 s worst case [D2, D5]	Met
R6	No mercury; full output in 0.1 s	LEDs, microsecond rise	Met
R7	Window 6.8 MPa or less at 8 bar	6.18 MPa [G4]	Met
R8	0.5 bar or less at 1.2 L/min, excluding the restrictor	0.15 bar on assumed Kv [G2]; 0.59 with a Kv 0.10 valve [G9]	At risk
R9	25 W flowing, 0.5 W standby	22.6 W, 0.23 W [E1, E2]	Met
R10	LED board 50 °C or less; cut-back above 50 °C	37.3 to 45.1 °C steady [F3]	Met
R11	Food-contact wetted parts; no UV-C on plastics but PTFE	316 lower cap; PTFE liner, top disc and 316 outlet sleeve shield the acetal upper cap; expanded PTFE window gasket (STEM fix 2026-10-25), EPDM O-rings out of the light	Met
R12	No UV-C outside the unit; interlocks	Metal and PTFE light path; sensor window sealed in the welded boss by the photodiode holder; interlocks: the lid reed switch, and an interlock loop in the LED head's plug, whose 95 mm cable is 39 mm short of letting the head clear the cap while plugged in (model check)	Met
R13	24 V DC only at the unit	Certified adapter and input fuse	Met
R14	350 x 150 x 350 mm or less	277 x 100 x 322 mm [H1]	Met
R17	Pressure limiter at 4 bar or less upstream	Window 6.18 MPa at twice the setting [G5]	Met

ID	Target	Value (tag)	Status
R18	Five-segment UV level bar, UV level relative to the alarm threshold	Three of five segments at 90 %/cm; last segment goes out where the alarm trips [C6]; bar in the model and BOM line 13	Met

Counts: 12 met, 3 at risk, 1 not met, 1 not verifiable at TRL 3, and R16 over its value-engineering target by USD 93 [K1].

Changes in v0.7 (kit 1.9.1 and the independent review, 2026-10-25)

Table 8. What the independent review of 2026-10-25 changed in this note.

Item	v0.6	v0.7
Organism response at 275 nm	Taken as equal to 254 nm	Factor 0.93 from published MS2 data at 280 nm: RED 47.3 mJ/cm² at 90 %/cm, 15.8 at 70 %/cm [B10]; R2 at risk
LED derating	20 % low output only [B9]	Stacked junction-temperature and bin derating with the 275 nm factor: 37.5 mJ/cm² [B11], to confirm on the chosen LED
Alarm threshold	S/S_ref 0.79 from the unweighted RED	0.85 from the weighted RED; one failed LED trips it [C3, C4]; the UV level bar shows two of five segments [C6]
Start-up	Flow sensor wakes the unit (impossible with the outlet valve shut)	Outlet pressure sensor wakes the unit; flow sensor confirms flow [D5]; standby 0.23 W [E2]
Pressure drop	Assumed Kv only	Kv 0.10 valve case 0.59 bar [G9]; R8 at risk
Cost	USD 416	USD 433: expanded PTFE gasket (USD 2) and outlet pressure sensor with tee (USD 15), both estimates [I1]

The review also showed that the laminar RED is not a strict lower bound: the profile never develops over the 226 mm flow path, the inlet jet and the warm lower cap mix the water, and a strongly back-mixed reactor would give less (LMF-STM-001, S34). The residence-time distribution is a TRL 4 measurement; an inlet flow distributor is proposed to Amish (LMF-DEC-001, O-4). The water-film coefficient in the lower cap, which the review questioned, is about 490 W/(m² K) for developing flow over the first 18 mm of stainless bore, above the 150 used for the hottest board case (S35).

Numbers changed from v0.1

Table 7. v0.1 (25 mm bore, 2.0 L/min) against v0.2 (LMF-DDR-002 design).

Quantity	v0.1	v0.2
Bore, channel, window	25 mm, 246 mm, 32 x 6 mm	50 mm, 242 mm, 57 x 10 mm [A1, G4]
Design flow	2.0 L/min	1.2 L/min [A3]
Liner reflectance (assumed)	0.80	0.95
UV-C absorbed by water at 90 %/cm	26.1 %	54.0 % [B3]
RED at 90 %/cm	14.7 to 19.0 mJ/cm² (R2 not met)	51.9 to 76.3 mJ/cm² (R2 met) [B5]
RED at 70 %/cm	6.3 to 7.4 mJ/cm²	17.4 to 21.8 mJ/cm² [B6]
Dose sensor signal at 90 %/cm and 70 %/cm	2.4 nA, unresolved	16.3 nA, 0.38 nA [C1]
Worst-case switch-on	0.46 s	0.24 s [D2]
Pressure drop	0.42 bar	0.15 bar [G2]
LED board, nominal	48.9 °C	40.3 °C [F3]
Window stress at 8 bar	4.46 MPa	6.18 MPa [G4]
Energy	3.1 kWh/year	3.8 kWh/year [E3]
Envelope	235 x 72 x 322 mm	277 x 100 x 322 mm [H1]
Parts cost	$249.00	$338.00 [I1]
Requirements	8 met, 4 at risk, 3 not met, 1 not verifiable (16)	13 met, 1 at risk, 2 not met, 1 not verifiable (17); 14 met and 1 not met in v0.3 after the budget top-up [K1]

Table 8. v0.3 against v0.4 (constructable design, LMF-DDR-003). Every other number is unchanged.

Quantity	v0.3	v0.4
Heat capacity of sink, ring, cap and board	802 J/K	781 J/K [F1]
Acetal lower cap case: time to 50 °C	31.4 min	30.5 min [F5]
Upper cap roof stress at 8 bar	not checked (3 mm roof, about 59 MPa)	2.8 MPa, 13 mm roof [G8]
Estimated cost	$338.00, 16 lines	$402.00, 18 lines [I1]
R16	Met	Over the value-engineering target by USD 62 [K1]

Table 9. v0.5 against v0.6 (decisions of 2026-10-02 carried into the design). Every other number is unchanged.

Quantity	v0.5	v0.6
LED to window gap	0.8 mm	1.3 mm, window on a 0.5 mm PTFE washer
UV-C entering the water	87.0 %	84.8 % [B2]
RED at 90 %/cm	51.9 to 76.3 mJ/cm²	50.9 to 74.9 mJ/cm² [B5]
RED at 70 %/cm	17.4 to 21.8 mJ/cm²	17.0 to 21.3 mJ/cm² [B6]
Flow for the 20 % margin at 90 %/cm	1.30 L/min	1.26 L/min [J1]
Wall sensor at 90 %/cm	16.3 nA	13.4 nA [C1]
Alarm threshold S/S_ref	0.77	0.79 [C3]
Window stress at 8 bar	6.18 MPa	6.18 MPa, unchanged [G4]
LED on-time	86 h/year	92 h/year with the idle pulse [E4, E6]
Mass	not estimated	4.97 kg dry, 5.44 kg full [H2]
Estimated cost	$402.00, 18 lines	$433.00, 19 lines [I1]
R18	Not verifiable at TRL 3	Met [C6]
Requirements	13 met	14 met, 1 not met, 1 at risk, 1 not verifiable, R16 over its target by USD 93 [K1]